

Report of the WHO meeting to review progress in implementing phase-2 of respiratory syncytial virus surveillance, based on the Global Influenza Surveillance and Response System

Virtual meeting, 18 and 21-22 June 2021



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Acknowledgements

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WHO acknowledges the contributions of all the national and international experts that participated in the meeting. Thanks are due to the Respiratory Syncytial Virus reference laboratories, the National Influenza Centres and the National Epidemiology Centres for their support to the Phase-2 implementations of the WHO Global RSV Surveillance project.

The report was prepared by Harry Campbell and reviewed internally by Siddhivinayak Hirve, Sandra Jackson, Christopher Chadwick and Wenqing Zhang from WHO and externally by Harish Nair (University of Edinburgh, United Kingdom of Great Britain and Northern Ireland) and Maria Zambon (UK Health Security Agency, United Kingdom), co-chairs for the meeting.

Acronyms

ARI	acute respiratory infection
COVID-19	coronavirus disease of 2019
EQAP	external quality assessment programme
GISRS	WHO Global Influenza Surveillance and Response System
GPW	WHO General Programme of Work
LMICs	low- and middle-income countries
RSV	respiratory syncytial virus
SARI	severe acute respiratory infection
SARS-CoV-2	severe acute respiratory syndrome-coronavirus 2

Background

Respiratory syncytial virus (RSV) is a leading cause of hospitalisations due to acute lower respiratory infections in infants and young children. Surveillance allows for the establishment of the evidence base that further guides the development of RSV vaccines and monoclonal antibodies and informs immunization policy and practice. In 2016-2018, WHO conducted a three-year (Phase-1) pilot project to test the feasibility of implementing RSV surveillance based on the Global Influenza Surveillance and Response System (GISRS) in 14 countries across all six WHO regions. Based on the findings, in April 2019, WHO updated the RSV surveillance strategy for Phase-2 of the project and implemented it in an additional 11 low and middle-income countries (LMICs). The RSV surveillance strategy is in alignment with and contributes to WHO's Thirteenth General Programme of Work (GPW-13) towards preparedness and prevention strategies for priority pandemic- and epidemic-prone diseases.

On 18, 21 and 22 June 2021, WHO hosted a virtual meeting to review progress in implementing Phase-2 of the RSV surveillance project. The meeting brought together 134 participants from all six WHO regions. Participants included laboratory and epidemiological surveillance focal points from the 25 countries participating in RSV surveillance; RSV reference laboratories; international RSV experts; global and regional partners and other interested bodies.

Scope and objectives

The overall aim of this meeting was to review the progress of implementation and challenges in RSV surveillance, identify solutions, and develop action plans to overcome the challenges and disruptions in the context of the ongoing COVID-19 pandemic.

The objectives of the meeting were to:

- review the progress in the implementation of RSV surveillance in 2019-2020;
- discuss and seek solutions to mitigate the disruptions in RSV surveillance in the context of the ongoing COVID-19 pandemic;
- review the performance of participating laboratories in the external quality assessment programme (EQAP);
- review the progress of RSV virus and sequence sharing, sequencing, and analytic pipelines for genetic sequence data;
- develop action plans for RSV surveillance for 2021–2022;
- review the expected project outcomes and sustainability proposals for RSV surveillance; and
- review and discuss experiences learnt from RSV surveillance for the potential development of GISRS towards a broader platform for influenza, RSV, SARS-CoV-2, and other respiratory viruses.

The expected outcomes of the meeting were:

- Measures to mitigate challenges and disruptions of RSV surveillance;
- Better understanding of RSV epidemiology and virology in the context of co-circulation with SARS-CoV-2 and influenza;
- Sharing of RSV genetic sequencing data in publicly accessible databases and analytic pipelines; and
- Action plans for RSV surveillance in 2021–22 in the context of the ongoing COVID-19 pandemic.

The following sections provide highlights from the meeting.

Introduction

During the meeting, WHO presented a review of Phase-1 (Pilot) achievements. This included the analysis of >33 000 specimens tested for RSV, the results of which were published in nine manuscripts in the *Influenza and Other Respiratory Viruses* journal supplement¹. Findings informed the development of Phase-2 of the project. Some key learnings from the pilot included:

- RSV surveillance as part of GISRS was feasible without adverse impacts on influenza surveillance;
- Extended severe acute respiratory infections (SARI) and acute respiratory infections (ARI) case definitions performed well for RSV, in particular with increased sensitivity for RSV;
- RSV seasonality was described in 14 participating countries; and
- RSV surveillance has been sustained over several years in some countries and such experience could be used as a template for national implementation.

Phase-2 findings highlighted ongoing RSV surveillance as critical going forward to understand epidemiology, seasonality, and age shifts; virologic evolution and interactions with SARS-CoV-2 and influenza co-circulation; and the impact of non-pharmacological interventions, vaccines, and monoclonal intervention strategies. The fields of monoclonal antibody and vaccine development continue to advance very rapidly, with likely more than one product to be available for licensure in the next two to three years. This emphasized the important need to maintain RSV surveillance and revitalise this project.

Achievements and challenges

¹ <https://www.who.int/teams/global-influenza-programme/global-respiratory-syncytial-virus-surveillance>

Although most participating countries experienced disruptions to systems impacting RSV surveillance, 80% (20/25) conducted routine or sporadic surveillance during 2020 with all countries reporting reductions in RSV circulation. Disruptions were most evident in new countries joining Phase-2 as many did not have time to set up adequately before the COVID-19 pandemic. Challenges included responding to COVID-19 demands, changes in healthcare seeking behaviours, increased workload on clinical services at sentinel sites, delays in reagent shipments, decreased availability/access to equipment, and increased staff turnover.

Among the countries engaged in surveillance activities in Phase-2, most had adopted the extended SARI and ARI case definitions and 16 were implementing year-round RSV surveillance. However, only nine countries had uploaded surveillance results to FluMart. Laboratory assays had been updated based on results from the Pilot. WHO facilitated access to reagents for all participating laboratories directly through WHO headquarters and the International Reagent Resource established by the Centers for Disease Control and Prevention in the United States of America. A standardized approach to quality assessment for the molecular diagnosis of RSV had been developed previously. All countries participated in the updated 2019-2020 WHO RSV EQAP and returned satisfactory results, indicating that results generated during Phase-2 will be reliable. RSV positive samples had been typed for A and B and RSV negative samples were tested for influenza. Participants uploaded 6 306 RSV test results (27% RSV positivity in <2 years old) and corresponding case data to FluMart as of 1 June 2021.

An RSV Genotyping Consensus Consortium has been established to propose a systematic approach to genotyping. The proposed RSV nomenclature to below sub-species level² was applied to Phase-2 isolates. Further work is needed for agreement on strain classification. Ten countries initiated sequencing with a further eight planning to start in 2021. All Phase-2 countries agreed to share sequences and adopt a common nomenclature. RSV reference laboratories reported back sequence data to countries and provided training and technical support to requesting countries.

The EpiRSV™ platform, which was developed by the GISAID Initiative and launched at the meeting, can be used for real-time genomic surveillance to study global trends and how mutations affect diagnostic kits and interventions. EpiRSV™ includes a curated database to achieve epidemiological and biological consistency and identify important variants, a bioinformatic pipeline to generate consensus sequences, and an EpiRSV™ reporting structure which records amino acid changes with respect to an agreed reference sequence. As of the launch, 23 365 virus sequences had been uploaded to GISAID EpiRSV™.

² Salimi, V, Viegas, M, Trento, A, Agoti, CN, Anderson, LJ, Avadhanula, V, et al. Proposal for Human Respiratory Syncytial Virus Nomenclature below the Species Level. *Emerg Infect Dis.* 2021;27(6):1-9. doi: 10.3201/eid2706.204608.

RSV surveillance

It was evident that participating countries remain committed to building and maintaining RSV surveillance on the GISRS platform. It is important for countries to indicate what support is needed to maintain surveillance both at present and in the future (in terms of programme sustainability) and to raise the profile of RSV with Ministries of Health and donors so that this work can be revitalised in 2021–2022.

There is a need at this time for year-round RSV surveillance, even in countries with robust historical seasonal pattern systems, although it was recognised that this was challenging during times when resources are stretched. In terms of estimating burden of disease, it is important to understand what percentage of eligible children are enrolled to estimate RSV burden in that facility and to have clarity on the representativeness of those enrolled; the addition of an objective measure of severity like hypoxemia may help to reduce variability in reporting across countries.

Integrated surveillance

There was a widely expressed desire to support countries towards developing integrated SARI sentinel systems that will serve needs for influenza and SARS-CoV-2 and RSV. A review of the case definitions to ensure that they perform well for these pathogens will therefore be necessary. Considerations of the experience in RSV surveillance from this project will be important in planning the future of GISRS surveillance, and it will be important to capitalise on the opportunities leading from the legacies of emerging coronaviruses for future integrated respiratory virus surveillance as a contribution to pandemic preparedness.

The design of an integrated surveillance system will depend on the priority objectives of the surveillance being addressed. This will require a wide range of views from many stakeholders to obtain a full view of public health and other needs. A broader GISRS+ platform was proposed with an integrated system for surveillance and monitoring of influenza, SARS-CoV-2, RSV, and other respiratory viruses. This would provide the laboratory and surveillance systems, and data to support global decision-making for vaccines, therapeutics, and diagnostics development and would continue to complement event-based surveillance for unusual events. This should build upon the existing GISRS infrastructure.

RSV reference laboratories

It was recommended that RSV reference laboratories should have a separate meeting to compare protocols and identify the most robust approaches; discuss sources of possible variation; discuss the EQAP; review reported problems with sequencing, and review Reference Laboratory workflow issues informed by results from the current EQAP analysis. All

laboratories should have access to reliable reference sequencing protocols soon so that progress can be made.

Sustainability

It was agreed that there is continuing need for data to inform vaccine and monoclonal antibody pipelines and to build capacity for surveillance platforms for future vaccine effectiveness study needs. An investment case for the continuation of RSV surveillance is needed to (1) make the case that further extension is needed to meet the original objectives at a time when RSV epidemiology has changed radically, (2) understand the indirect effects of the pandemic and removal of non-pharmaceutical interventions. It was noted that additional opportunities now exist to leverage the enthusiasm and attention given to respiratory viruses in the past two years. It will be necessary to capitalise on this to seek governmental and stakeholder support and funding for future surveillance beyond the current project.

Future directions

Phase-2 was intended to end on 31 October 2021; this has since been extended by 12 months due to COVID-19 pandemic delays. Several critical actions will be needed in the next 12 months to advance RSV surveillance and meet specific project deliverables. It was agreed that a detailed understanding of country concerns about the proposed critical actions in surveillance is necessary since it is important to have surveillance systems in place before the next RSV season.

Close of meeting

The meeting was concluded by Dr Wenqing Zhang (WHO/GIP) who acknowledged the commitment and efforts of participants and congratulated them on the important progress made despite many challenges. The significant developments, which have contributed towards the improvement of public health capacity to control the high disease burden of RSV, were noted. The next steps include continued implementation of the Phase-2 strategy, which describes a detailed path forward to achieve the remaining project objectives.

Annex 1: Agenda

WHO Meeting to Review Progress in Phase-2
of the RSV Surveillance based on the GISRS
(Virtual Meeting)

AGENDA
(Meeting room will open at 10h30 CET each day)

Friday, 18 June 2021 (Day 1) 11h00 – 14h30 CET Co-chairs: Harish Nair / Maria Zambon

11:00 – 11:10	Opening Appointment of co-chairs, rapporteur	Wenqing Zhang
Session 1 – Looking back		
11:10 – 11:25	WHO RSV surveillance – overview and progress - Q&A	Siddhi Hirve
11:25 – 11:40	Progress status – laboratory - Q&A	Sandra Jackson
11:40 – 12:00	RSV seasonality – what lies ahead? - Q&A	Dan Weinberger
Session 2 – Sharing experiences		
12:00 – 12:05	Introduction of breakout sessions (5 min)	Siddhi Hirve
12:05 – 13:35	Country experiences (room #1) Central African Republic, Côte d’Ivoire, Madagascar, Mozambique, Senegal, South Africa, Uganda	Session chair (room #1) Marietjie Venter
	Country experiences (room #2) Argentina, Brazil, Canada, Chile, Russian Federation, United Kingdom	Session chair (room #2) Pasi Penttinen
	Country experiences (room #3) Egypt, Jordan, Lebanon, Morocco, Pakistan, Qatar	Session chair (room #3) Peter Smith
	Country experiences (room #4) India, Nepal, Thailand, Australia, Mongolia, Philippines	Session chair (room #4) Shobha Broor
13:35 – 13:45	Health break	
13:45 – 14:25	Plenary – Summarizing regional experiences - Q&A	Belinda Herring (room #1) Juliana Leite / Richard Pebody (room #2) Lubna Al Ariqi (room #3) Karen Nahapetyan / Francis Inbanathan (room #4)
14:25 – 14:30	Wrap up	Co-chairs

Monday, 21 June 2021 (Day 2) 11h00 – 14h00 CET

11:00 – 11:10 Recap of Day 1

Co-chairs: *Harish Nair / Maria Zambon*

Session 3A – (Epi)

Chair: *Harish Nair*; Rapporteur: *Harry Campbell*

11:10 – 11:40	Discussion: Revitalizing RSV sentinel surveillance	Moderator: <i>Meredith McMorrow</i>
11:40 – 12:15	Discussion: RSV in the context of COVID-19 – using surveillance data to inform interventions and policy	Moderator: <i>John Paget / Danny Feikin</i>
12:15 – 12:30	Measuring RSV hospitalization burden	<i>Jocelyn Moyes</i>
12:30 – 12:40	Health break	
12:40 – 12:50	Launch of EpiRSV™	<i>Peter Bogner</i>
12:50 – 13:20	EpiRSV™ – what's in the toolbox?	<i>Sebastian Maurer-Stroh</i>
13:20 – 13:50	RSV molecular epidemiology- preliminary results and implications of RSV surveillance	<i>Ian Barr</i> <i>Everardo Vega</i> <i>Marilda Sequeira</i> <i>Almiro Tivane</i>
13:50 – 14:00	Wrap up	Co-chairs

Session 3B – (Lab)

Chair: *Maria Zambon*; Rapporteur : *Rodrigo Fasce*

11:10 – 11:25	WHO RSV EQA 2019-2020	<i>Sandra Jackson Thomas Williams</i> <i>Sanjiv Rughooputh</i> <i>Brett Whitaker</i>
11:25 – 11:40	Phase-2 overview. sample sharing, nomenclature, subtyping, sequencing, training needs	Moderator: <i>Ian Barr</i>
11:40 – 11:55	The RSV genome: RSV F, G and WGS Phenotype relationships	<i>Thomas Williams</i>
11:55 – 12:10	CDC: Reagents, sequencing algorithm pipeline (Vpipe) CDC workplan RSV Reference Lab updates	<i>Everardo Vega</i>
12:10-12:30	Discussion	<i>Joanna Ellis</i> <i>Anne von Gottberg</i>

Tuesday, 22 June 2021 (Day 3) 11h00 – 14h00 CET

11:00 - 11:10	Recap of Day 2	Co-chairs: Harish Nair / Maria Zambon
11:10 – 11:50	Plenary - Epi feedback; Q&A - Lab feedback; Q&A	Harry Campbell Rodrigo Fasce
Session 4 – Looking ahead		
11:50 – 12:05	WHO perspectives – GISRS+ including RSV	Ann Moen
12:05 – 12:20	BMGF perspectives – Investing in RSV, SARS-CoV-2 and ORV surveillance - Q&A	Padmini Srikantiah
12:20 – 12:30	Health break	
12:30 – 13:35	Implementing RSV surveillance in the current context - 2021–2022 workplan - Final deliverables of the project – what, how, who? - Discussion	Sandra Jackson Siddhi Hirve
		Moderator: Harish Nair
13:35 – 13:45	Next steps	Wenqing Zhang
13:45 – 13:55	Summary from Co-chairs	Harish Nair / Maria Zambon
13:55 – 14:00	Summary and closing	Ann Moen
14:00	Closure of meeting	

Annex 2: List of Participants

- Abdul Ahad, National Institutes of Health, Pakistan
- Adam MacNeil, Centers for Disease Control and Prevention, United States of America
- Ahmed Rguig, Ministry of Health, Morocco
- Ali Ben Hadj Kacem Mohamed, National Influenza Center, Qatar
- Almiro Tivane, National Influenza Center, Mozambique
- Amary Fall, Institute Pasteur, Senegal
- Amel Hasanin Naguib, National Influenza Center, Egypt
- Ammar Aziz, Royal Melbourne Hospital, Australia
- Anderson Ngattia, National Institute of Public Health, Côte d'Ivoire
- Annabelle Sucuano, Research Institute for Tropical Medicine, Philippines
- Anne von Gottberg, National Institute for Communicable Diseases, South Africa
- Annette Alafaci, Murdoch Children's Research Institute, Australia
- Antso H. Raherinandrasana, Ministère de la Santé Publique, Madagascar
- Aron Hall, Centers for Disease Control and Prevention, United States of America
- Audrey Dubot-Pérès, Emerging Viral Diseases Unit (UVE), France
- Bakamutumaho Barnabas, Uganda Virus Research Institute, Uganda
- Boly Diop, Ministry of Health and Social Action, Senegal
- Braulia Caetano, Fundação Oswaldo Cruz (FIOCRUZ), Brazil
- Brett Whitaker, Centers for Disease Control and Prevention, United States of America
- Carla Voto, Ministry of Health, Argentina
- Catherine Dacasin, Research Institute for Tropical Medicine, Philippines
- Celine Gurry, GISAID, Germany
- Chowdhary Lokbandhu, National Public health Laboratory, Nepal
- Christina Bancej, Public Health Agency, Canada
- Conall Watson, Public Health England, United Kingdom
- Daniel Weinberger, Yale School of Public Health, United States of America
- Daouda Coulibaly, National Institute of Public Health, Côte d'Ivoire
- David Blazes, Bill & Melinda Gates Foundation, United States of America
- Deborah Higgins, PATH, United States of America
- Eeva Broberg, GISAID, Sweden
- Elizaveta Smorodintseva, Research Institute of Influenza, Russian Federation
- Elsa Baumeister, INEI-ANLIS "Carlos G. Malbrán", Argentina
- Eric Simoes, Center for Global Health, Colorado, United States of America
- Erik Karlsson, Institute Pasteur, Cambodia
- Everardo Vega, Centers for Disease Control and Prevention, United States of America
- Felipe Cotrim de Carvalho, Saúde Brasil, Brazil
- Gordon Mpamize, Joint Clinical Research Centre, Uganda
- Gunter Bach, GISAID, Germany
- Hala Abou Naja, Ministry of Health, Lebanon
- Harish Nair, The University of Edinburgh, United Kingdom
- Harry Campbell, The University of Edinburgh, United Kingdom

- Herve Kadjo, Institute Pasteur, Côte d'Ivoire
- Hicham Oumzil, Ministry of Health, Morocco
- Ian Barr, Victorian Infectious Diseases Reference Laboratory, Australia
- Jean-Michel Heraud, Institute Pasteur, Madagascar
- Jinal Bhiman, The National Institute for Communicable Diseases, South Africa
- Joanna Ellis, Public Health England, United Kingdom
- Jocelyn Moyes, National Institute for Communicable Diseases, South Africa
- Joelinotahina H. Rabarison, Institute Pasteur, Madagascar
- John Paget, Netherlands Institute for Health Services Research (NIVEL), Netherlands
- Jonjee Morin, Research Institute for Tropical Medicine, Philippines
- Joyce Namulondo, Virus Research Institute, Uganda
- Jyoti Acharya, National Public Health Laboratory, Nepal
- Kim Mulholland, Murdoch Children's Research Institute, Australia
- Koshal Chandra Subedi, Ministry of Health and Population, Nepal
- Lien Anh Ha Do, Murdoch Children's Research Institute, Australia
- Lok Bandhu Chaudhary, National Public Health Laboratory, Nepal
- Louis Bont, Universitair Medisch Centrum Utrecht, Netherlands
- Mahmoud Yacoub, Centers for Disease Control and Prevention, Jordan
- Manal Morcos Fahim, Ministry of Health, Egypt
- Mara Russo, Saúde Brasil, Brazil
- Marcela Santos Correia da Costa, Saúde Brasil, Brazil
- Maria Fernanda Olivares, Ministry of Health, Chile
- Maria Pisareva, Research Institute of Influenza, Russian Federation
- Maria Zambon, Public Health England, United Kingdom
- Mariana Viegas, National Scientific and Technical Research Council, Argentina
- Marie Vernet, Institute Pasteur, Central African Republic
- Marietjie Venter, National Institute for Communicable Diseases, South Africa
- Marilda Siqueira, Fundação Oswaldo Cruz (FIOCRUZ), Brazil
- Mayan Lumandas, Research Institute for Tropical Medicine, Philippines
- Mend Tsogt, National Center for Communicable Diseases, Mongolia
- Meredith McMorrow, Centers for Disease Control and Prevention, United States of America
- Miquelina Chicanequisso Vaz, Ministry of Health, Mozambique
- Mirela Pale, Instituto Nacional de Saúde, Mozambique
- Miriam Terezinha, Saúde Brasil, Brazil
- Mohamed Ali, Laboratory Medicine and Pathology, Qatar
- Mohannad Ramadan, School of Medicine, Jordan
- Mona Albeaini, Rafic Hariri University Hospital, Lebanon
- Naema Al-Mawlawi, Laboratory Medicine and Pathology, Qatar
- Naomi Komadina, Victorian Infectious Diseases Reference Laboratory, Australia
- Ndongo Dia, National Influenza Center, Senegal
- Neuza Nguenha, National Influenza Center, Mozambique
- Nigel Crawford, SAEFVIC - Murdoch Children's Research Institute, Australia
- Niteen Wairagkar, United States of America

- Norosoa Razanajatovo, Institute Pasteur, Madagascar
- Oiythip Yasopa, Ministry of Health, Thailand
- Padmini Srikantiah, Bill & Melinda Gates Foundation, United States of America
- Paola Resende, Fundação Oswaldo Cruz (FIOCRUZ), Brazil
- Pasi Penttinen, European Centers for Disease Control and Prevention, Sweden
- Peter Bogner, GISAID, Germany
- Peter George Smith, London School of Hygiene & Tropical Medicine, United Kingdom
- Philippe Dussart, National Influenza Center, Madagascar
- Pilailuk Okada, National Influenza Center, Thailand
- Rahombanjanahary Norosoa, Institute Pasteur, Madagascar
- Raphael Tze Chuen Lee, A-Star-Education, Singapore
- Rodrigo Fasce, National Influenza Center, Chile
- Runa Jha, National Public Health Laboratory (NPHL), Nepal
- Saleem Wedad, National Influenza Center, Qatar
- Sana Tamim, National Institute of Health, Pakistan
- Sanjiv Rughooputh, Public Health England (PHE), United Kingdom
- Sebastian Maurer-Stroh, GISAID, Singapore
- Shabana Bi, Public Health England (PHE), United Kingdom
- Shobha Broor, India
- Silke Buda, The Robert Koch Institute, Germany
- Sonia M Raboni, Universidade Federal do Paraná, Brazil
- Stephanie Goya Ricardo Gutiérrez, Children's Hospital, Argentina
- Sumit Bhardwaj, ICMR-National Institute of Virology Pune, India
- Suthida Muangnoicharoen Hearn, Ministry of Health, Thailand
- Suzanne Elgohari, Public Health England, United Kingdom
- Sylvie van der Werf, Institute Pasteur, France
- Teresa Peret, Centers for Disease Control and Prevention, United States of America
- Theodor Ziegler, Finland
- Thomas William, University of Edinburgh, United Kingdom
- Thulisa Mkhencele, National Institute for Communicable Diseases, South Africa
- Tiina Talts, Public Health England, United Kingdom
- TS. Naranzul, National Center for Communicable Diseases, Mongolia
- Tsiry H. Randriambolamanantsoa, Institute Pasteur, Madagascar
- Varsha Potdar, National Influenza Center, India
- Vina Lea Arguelles, Research Institute for Tropical Medicine, Philippines
- Walquiria Aparecida Ferreira de Almeida, Saúde Brasil, Brazil
- Walter Haas, The Robert Koch Institute, Germany
- Wedad Saleem, Laboratory Medicine and Pathology, Qatar
- Wei Cai, The Robert Koch Institute, Germany
- Whitney Skowronski, Centers for Disease Control and Prevention, United States of America
- Xiaomin Dong, Victorian Infectious Diseases Reference Laboratory, Australia
- Xiaoyan Lu, Centers for Disease Control and Prevention, United States of America
- Yi-Mo Deng, Victorian Infectious Diseases Reference Laboratory, Australia

WHO Secretariat

Regional Office for Africa

Belinda	Herring	Emergency Preparedness
---------	---------	------------------------

Regional Office for the Americas

Andrea S.	Vicari	Infectious Hazard Management
Angel	Rodriguez	Infectious Hazard Management
Jorge	Jara	Infectious Hazard Management
Juliana	Leite	Infectious Hazard Management
Paula	Couto	Infectious Hazard Management

Regional Office for the Eastern Mediterranean

Abdinasir	Abubakar	Infectious Hazard Prevention and Preparedness
Amal	Barakat	Infectious Hazard Prevention and Preparedness
Lora	Alsawalha	WHO Representative's Office Jordan
Lubna	Al Ariqi	Infectious Hazard Prevention and Preparedness
Moubadda	Assi	WHO Representative's Office Lebanon
Uzma	Bashir	WHO Representative's Office Pakistan

Regional Office for Europe

Cyril	Barbezange	Infectious Hazard Management
Richard	Pebody	Infectious Hazard Management

Regional Office for South-East Asia

Arunkumar	Govindakarnavar	WHO Representative's Office Nepal
Francis Y.	Inbanathan	Infectious Hazard Management
Nirajan	Bhusal	WHO Representative's Office Nepal
Priya	Jha	WHO Representative's Office Nepal
Saugat	Shrestha	WHO Representative's Office Nepal

Regional Office for the Western Pacific

Karen	Nahapetyan	Infectious Hazard Management
Rowena	Capistrano	WHO Representative's Office Philippines

WHO Headquarters

Adam	Cohen	Essential Programme on Immunization
Ann	Moen	Influenza Preparedness and Response
Aspen	Hammond	Global Influenza Programme
Awandha	Mamahit	Global Influenza Programme
Bikram	Maharjan	Global Influenza Programme
Christian	Fuster	Global Influenza Programme
Daniel	Feikin	Essential Programme on Immunization
Dmitriy	Pereyaslov	Global Influenza Programme
Elizabeth	Frodeman	Global Influenza Programme

Erin	Sparrow	Product & Delivery Research
Hassene	Debbiche	Work Place and Collaboration Services
Hien	Doan	Global Influenza Programme
Julia	Fitzner	Global Influenza Programme
Katelijan	Vandemaele	Global Influenza Programme
Magdi	Samaan	Global Influenza Programme
Maja	Lievre	Global Influenza Programme
Marie-jo	Medina	Global Influenza Programme
Martin	Friede	Product & Delivery Research
Sandra	Jackson	Global Influenza Programme
Siddhivinayak	Hirve	Global Influenza Programme
Sylvie	Briand	Infectious Hazard Management
Vanessa	Cozza	Global Influenza Programme
Wenqing	Zhang	Global Influenza Programme

